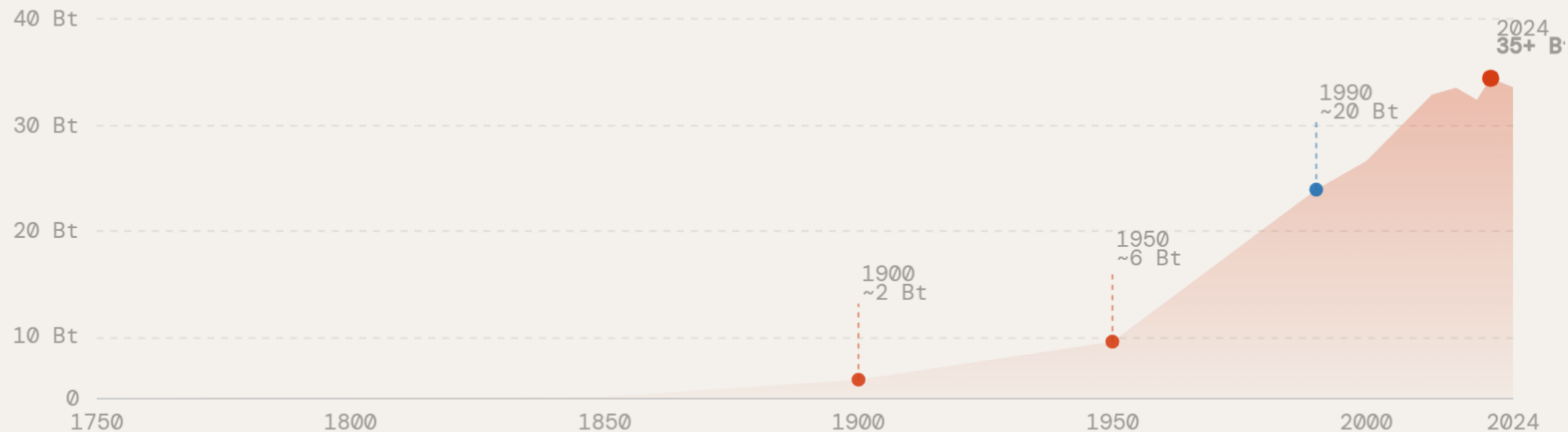


Global CO₂ Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry emissions from 1750 to 2024 — trajectories, regional shifts, and per capita inequalities informing climate responsibility.

Source — Global Carbon Budget 2025 via Our World in Data
Authors — Hannah Ritchie & Max Roser
Audience — Policy analysts, climate negotiators, sustainability researchers

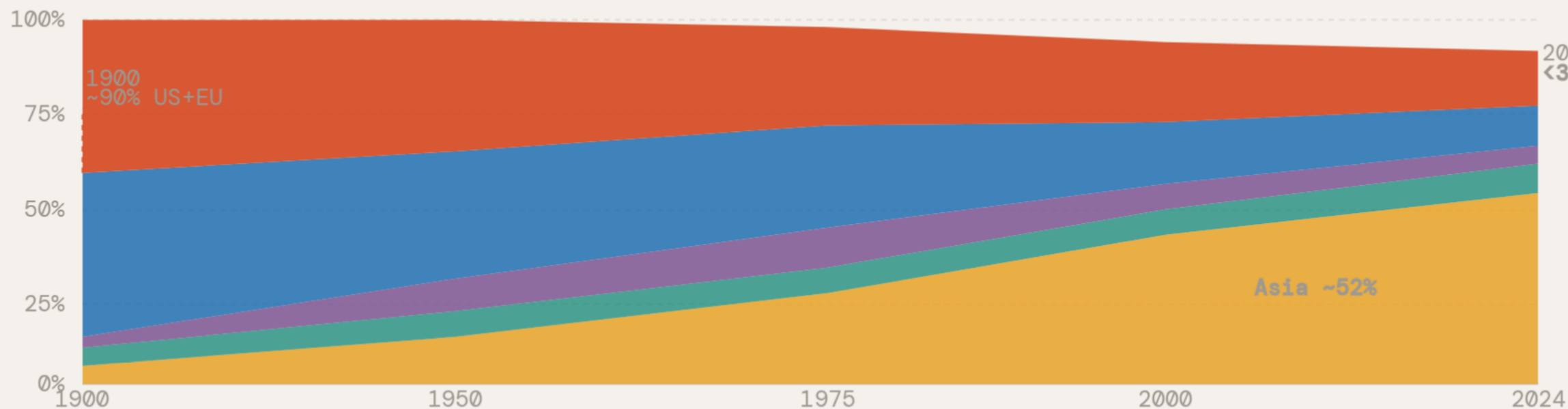
Global CO₂ Emissions Grew from **6 Bt** in 1950 to Over **35 Bt** Today



x5.8
GROWTH SINCE 1950

Growth has slowed since 2015, but **global emissions have not yet peaked**. Land-use change emissions have declined slightly; fossil fuel emissions continue to dominate.

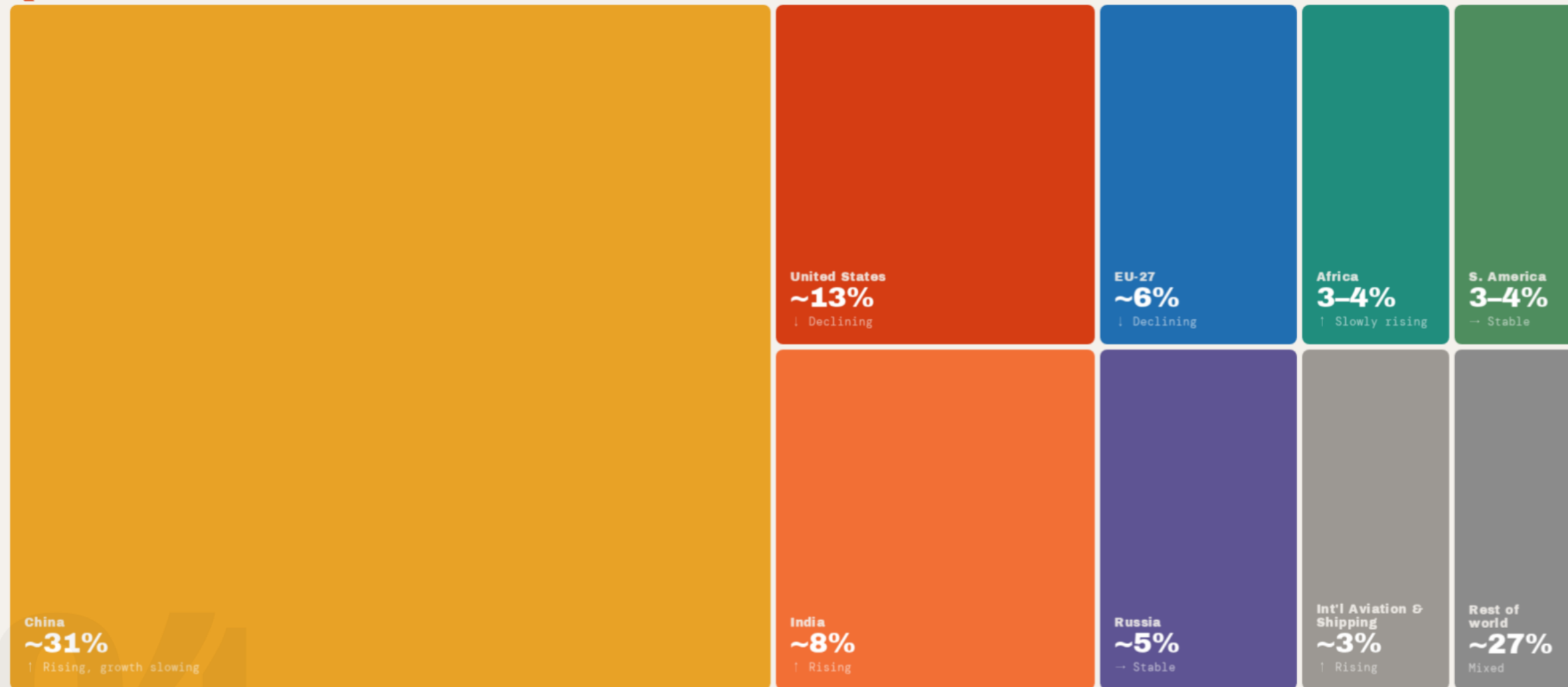
Europe & the US Fell from **90%** of Emissions in 1900 to Under **One-Third** Today



US & Canada Europe Russia / FSU Other developing Asia (China, India, other)

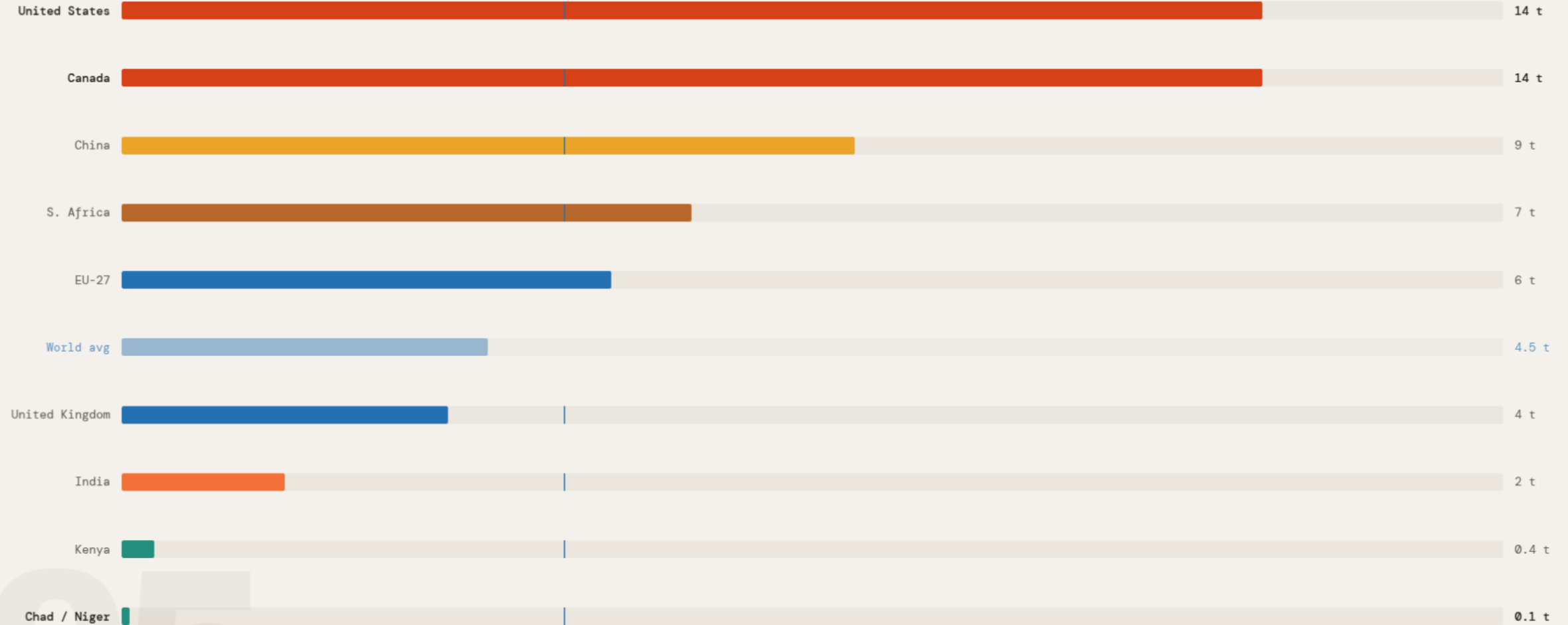
Asia — led by China — now produces roughly half of global emissions, up from under 5% in 1900. Europe and North America fell from >90% to under one-third over the same period.

China Now Emits More Than **One-Quarter** of Global CO₂



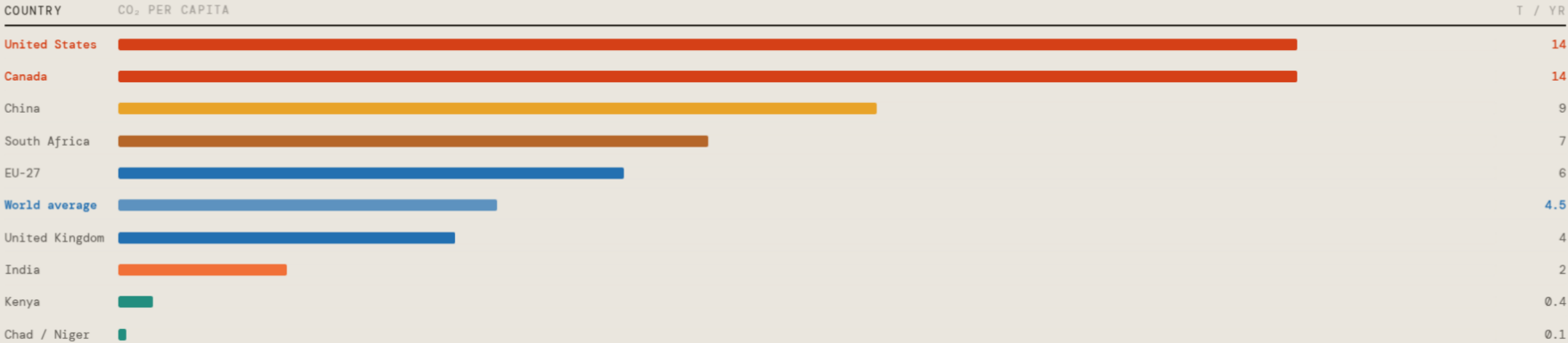
Cell area proportional to share of global CO₂ emissions. Africa and South America combined ≈ international aviation and shipping.

Per Capita Emissions Range from **0.1 t** in Chad to **~14 t** in the US and Canada



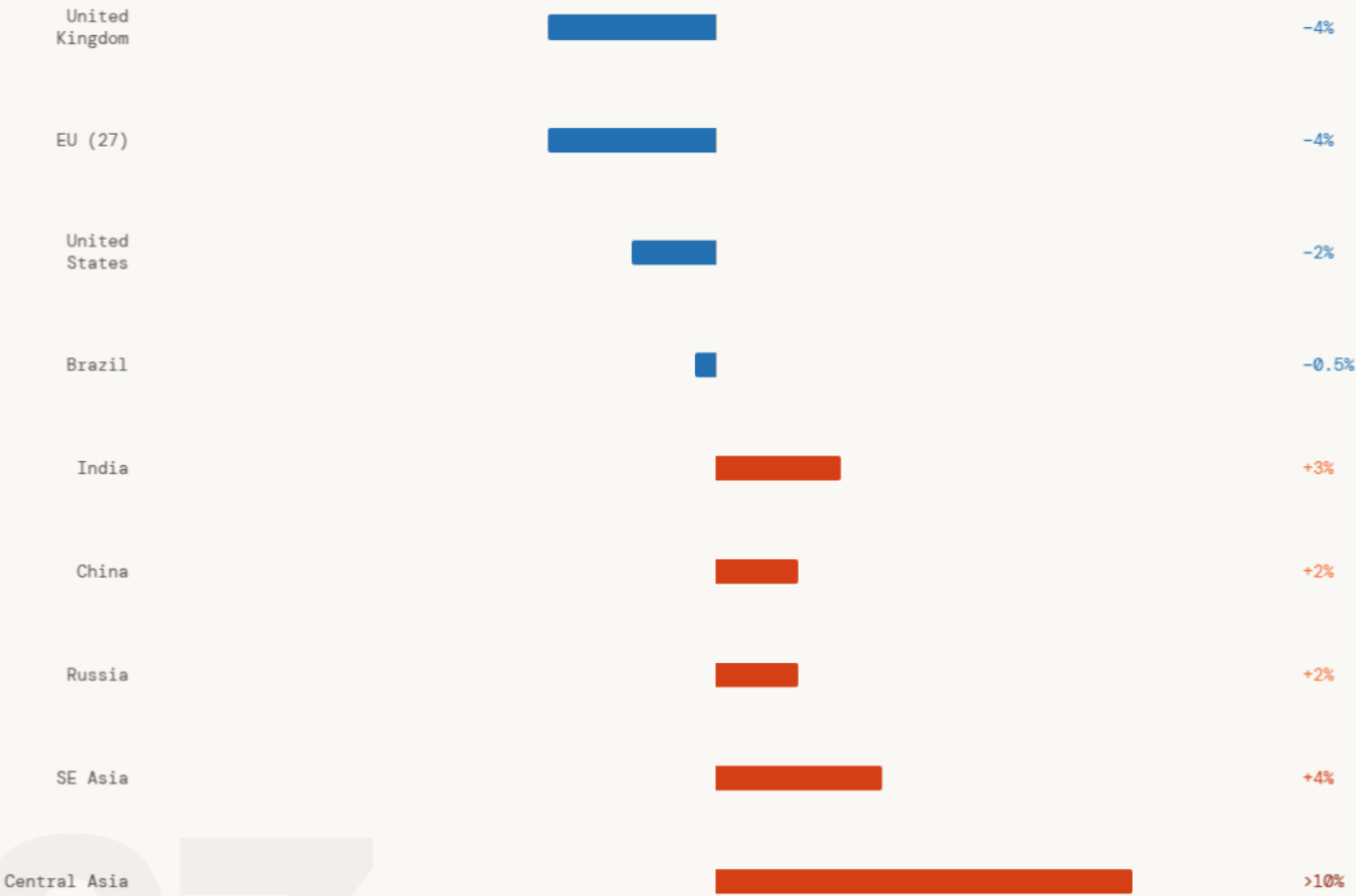
The 150x gap between Chad/Niger (0.1 t) and the US/Canada (14 t) means an average American emits in under two days what an average person in Mali emits in a year. [Blue line](#) = world average (4.5 t).

Per Capita Emissions: ~14 t in the US vs. ~9 t in China and ~2 t in India



China is the world's **largest total emitter** (~31%) but its per capita footprint (9 t) remains well below the US (14 t). India's per capita emissions (2 t) remain very low despite rapid growth.

Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



Declining emissions (blue)
Increasing emissions (red/orange)

Many developed nations reduced emissions in 2024, but increases in **Central Asia, SE Asia, and parts of Africa** offset some of these gains. The net global total remains above 35 Bt.

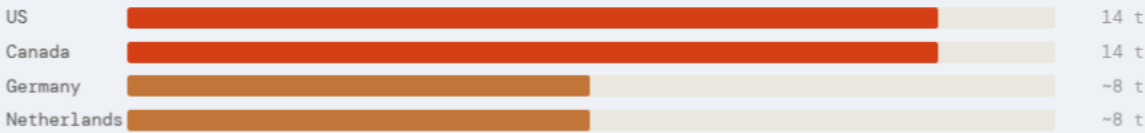
Note: Direction and approximate magnitude from Global Carbon Budget 2025. Central Asia includes Turkmenistan / Uzbekistan region with >10% annual increase.

Decrease

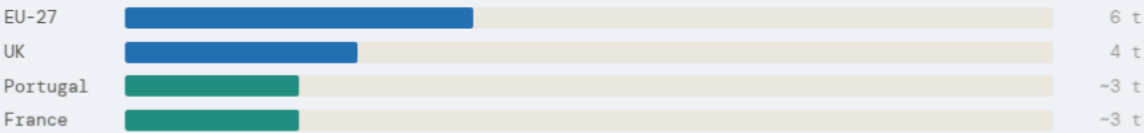
Increase

Energy Mix Explains Why France & the UK Emit Far **Less** Per Capita Than Germany or the US

HIGHER FOSSIL-FUEL SHARE → HIGHER PER CAPITA



HIGHER NUCLEAR & RENEWABLE SHARE → LOWER PER CAPITA



Coal and gas-heavy grids keep per capita figures elevated — comparable prosperity, higher carbon cost.

France (nuclear) and Portugal (renewables) achieve similar prosperity with **3–5× lower per capita emissions** than the US — demonstrating that energy policy, not just income, drives the gap.

Per capita figures from Global Carbon Budget 2025 · Germany/Netherlands/Portugal estimated from EU-27 range and qualitative direction stated in source.

KEY TAKEAWAYS

35+

BT CO₂ PER YEAR

150x

PER CAPITA GAP

Responsibility Is **Shared** but **Unequal**

- **No peak yet:** Global emissions exceed 35 Bt/yr — growth has slowed but not reversed.
- **Largest emitter vs. highest per capita:** China leads total share (~31%) but per capita (9 t) remains well below the US (14 t).